CS4048-Data Science

Marks 100 **Assignment No 2** **Due Date: March 3, 2026**

Q1. Implement Simple Linear Regression and Polynomial (Degree 2) Regression on the given dataset 1 and dataset 2 and compare the results. Find Regression Equations. Draw Scatter Plot as well. Calculate R^2 as well. Submit solution by (hand/doc/word) and in Python. **(25marks)**

X	Y
3	2.5
4	3.2
5	3.8
6	6.5
7	11.5
8	14.6

X	Y
1	2
2	4
3	9
4	16

Q2. Calculate Polynomial Regression Equation (Quadratic) degree 2 for the following data in Python: **(25 marks)**

Years of experience	Salary ($)
1	50000
2	55000
3	65000
4	80000
5	110000
6	150000
7	200000

Q3. Implement simple linear regression using Sklearn in Python. You are required to import the dataset into your file and perform data pre-processing steps. Use 75% of the total dataset in the training of your model. The screenshot of first twenty rows of your dataset "titanic.csv" is attached for your reference. The dataset might have some missing values but Fare for each ticket is given. Your goal is to predict whether the passenger survived the sinking of the Titanic or not. Moreover, you are required to plot the best fit line on your dataset. You can remove unnecessary features if any. **(25 marks)**

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, M	male	22	1	0	A/5 21171	7.25		S
2	1	1	Cumings, N	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen,	female	26	0	0	STON/O2.	7.925		S
4	1	1	Futrelle, M	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr.	male	35	0	0	373450	8.05		S
6	0	3	Moran, Mr	male		0	0	330877	8.4583		Q
7	0	1	McCarthy,	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, M	male	2	3	1	349909	21.075		S
9	1	3	Johnson, N	female	27	0	2	347742	11.1333		S
10	1	2	Nasser, Mr	female	14	1	0	237736	30.0708		C
11	1	3	Sandstrom	female	4	1	1	PP 9549	16.7	G6	S
12	1	1	Bonnell, M	female	58	0	0	113783	26.55	C103	S
13	0	3	Saunderco	male	20	0	0	A/5. 2151	8.05		S
14	0	3	Andersson	male	39	1	5	347082	31.275		S
15	0	3	Vestrom, N	female	14	0	0	350406	7.8542		S
16	1	2	Hewlett, N	female	55	0	0	248706	16		S
17	0	3	Rice, Mast	male	2	4	1	382652	29.125		Q
18	1	2	Williams, N	male		0	0	244373	13		S
19	0	3	Vander Pla	female	31	1	0	345763	18		S
20	1	3	Masselmai	female		0	0	2649	7.225		C

Q4. Show the steps of Apriori and FPGrowth Algorithm on the following data. There are 9 items A1 to A9 (9 columns) and 15 transactions (rows). The support count is 3. **(25 marks)**

A1	A2	A3	A4	A5	A6	A7	A8	A9
1	0	0	0	1	1	0	1	0
0	1	0	1	0	0	0	1	0
0	0	0	1	1	0	1	0	0
0	1	1	0	0	0	0	0	0
0	0	0	0	1	1	1	0	0
0	1	1	1	0	0	0	0	0
0	1	0	0	0	1	1	0	1
0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	1	0
0	0	1	0	1	0	1	0	0
0	0	1	0	1	0	1	0	0
0	0	0	0	1	1	0	1	0
0	1	0	1	0	1	1	0	0
1	0	1	0	1	0	1	0	0
0	1	1	0	0	0	0	0	1